Tyler Chisholm

Technical Designer · Real-Time Rendering

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Technical designer with a rendering focus. I've built a hardware ray-traced Vulkan renderer (ReSTIR, SVGF, DLSS Ray Reconstruction) that holds 90 Hz in VR, and shipped UE5 features on a team (compute-shader flocking, lighting architecture, Chaos destruction, editor tools). I profile before I optimize.

Education

M.S. Interactive Entertainment, Florida Interactive Entertainment Academy (FIEA), University of Central Florida. Dec 2026
12 games built across 4 semesters in UE5, Unity and VR.

B.B.A. Finance & Entrepreneurship, Stetson University, DeLand, FL May 2024

Skills

Rendering	Vulkan ray tracing, ReSTIR, SVGF denoising, ray-traced GI, PBR, volumetrics, compute shaders, raymarching/SDFs, stencil techniques
Engines	Unreal Engine 5 (Nanite, Lumen, VSM, Substrate, Niagara, Chaos), Unity URP, Godot (engine fork)
Languages	HLSL, GLSL, C++, C#, Rust, Python
Tools	Nsight Graphics, Perforce, Git, Blender, Unreal Python, Jira

Rendering

Spatial Engine — ray-traced engine for non-Euclidean spaces

Rust · Vulkan RT · HLSL · OpenXR 2026 – Present

Personal engine, ~115k lines of Rust and 40k of HLSL across 13 crates, 680+ commits. I own the architecture, direction and verification; implementation is AI-assisted.

- Hardware ray-traced Vulkan renderer where spaces with different geometry (hyperbolic, spherical, toroidal) connect through portals, with lighting and shadows traced across them.
- Runs the full path tracer in VR at 90 Hz on a Quest over Link on modest RTX hardware, across ~900 km of streamed terrain with volumetric clouds.
- Lighting stack: ReSTIR direct lighting, one-bounce diffuse GI, SVGF denoising, physically based sky and volumetric atmosphere, DLSS Ray Reconstruction; analytic intersection shaders for curved primitives.
- Found and fixed a shader performance regression from GPU timestamp traces by replacing a 64-step sphere tracer with an analytic quadratic solve.
- Integrated into a Godot 4.6 fork as a custom renderer behind a C ABI, so Godot supplies the editor, scripting and asset pipeline.

Design Lead — Everent (Studio Kismet, FIEA capstone)

UE5.7 · C++ · Blueprint · Chaos · Perforce 2025 – Present

Momentum-based hoverboard game across an open world of floating sky islands in UE5.7 (Nanite, Virtual Shadow Maps, Substrate, World Partition), coming to Steam.

- Designed and built speed-gated destruction in C++ on Chaos Geometry Collections: a predictive sweep measures impact speed into the wall, drops blockers a frame early to keep momentum, and fires strain, directional and radial velocity fields. It doubles as secret gating and skill expression.
- Built the Destructible Wall Browser editor tool: lists every destructible in the level, exposes its tuning and fires a test projectile at a set point and speed, so designers tune without entering Play mode.
- Prototyped a compute-shader GPU flocking plugin for ambient sky creatures; when Niagara integration failed, pivoted to instanced meshes driven by CPU readback.
- Led the lighting and shadow architecture decision (Lumen vs. baked GI, VSM cost, HLOD) and wrote the team's graphics-settings spec on UGameUserSettings with runtime scalability groups.

Selected Projects

Pocket Spaces — stencil portal system (bigger on the inside)

Unity URP · C# 2026

- Doorways reveal rooms larger than their exteriors via stencil masking; a player state machine, per-object portal membership and gravity overrides let the player and physics objects cross cleanly.
- Raymarched hyperbolic Mandelbox volume inside a pocket room with Möbius-transform camera movement; documented stencil depth limits (render queue collapse at depth 3+).

Earlier Experience

Quality Assurance Agent — Stormbreaker Studios, Orlando, FL

Jan – Jul 2020

Tested 8 iOS client apps; wrote 85 defect tickets with repro steps and fix recommendations.